ChemCritic 文献证据卡

PMC11731305 | C – H 活化 / 芳基化 | 采集状态 imported_unchecked

标题	TiO2-Catalyzed Direct Diazenylation of Active Methylene Compounds with Diazonium Salts
作者	Evangelista E., de Carvalho Y., de Jesus I., Rodrigues M., Hayashi E., de Andrade K. 等 8 人
出处	The Journal of Organic Chemistry 2024
DOI	10.1021/acs.joc.4c02266
原文链接	https://pmc.ncbi.nlm.nih.gov/articles/PMC11731305/
许可	https://creativecommons.org/licenses/by/4.0/Permits the broadest form of re-use including for commercial purposes, provi

摘要（原文节选）

Herein, we report a direct diazenylation of active methylene compounds that is environmentally friendly, cost-effective, and scalable, utilizing a heterogeneous TiO2 catalyst that is both accessible and recyclable. The scope of this transformation shows excellent compatibility with both electron-rich and electron-poor diazonium salts, yielding the desired products in very good yields at room temperature. ⋯

数据表 · Table 1（11 行，含数值行 11 行）

Optimization of the Reaction Conditions

entry	catalyst	solvent	time (h)	temperature	light irradiation	yield (%)
1	TiO2 (0.25 equiv)	Et0H	0.5	rt		81
2	TiO2 (0.25 equiv)	H2O	0.5	rt		52
3	TiO2 (0.25 equiv)	Me0H	0.5	rt		57
4	TiO2 (0.25 equiv) CH3CN	CH3CN	0.5	rt		67
5	Al2O3 (0.25 equiv)	Et0H	0.5	rt		54
6	Cu2O (0.25 equiv)	Et0H	0.5	rt		40
7	TiO2 (1.0 equiv)	Et0H	0.5	rt		88
8	TiO2 (1.0 equiv)	Et0H	1	rt		70
9	absence	Et0H	48	rt		60
10	TiO2 (1.0 equiv)	Et0H	1	rt	blue LED	61
11	TiO2 (0.25 equiv)	EtOH	1	rt		

正文对应句（逐字）：We started our study with a preliminary investigation of the reaction between 4-hydroxycoumarin 1 and aryldiazonium salt 2a to evaluate the effect of various experimental parameters on the reaction outcome. After a set of preliminary experiments, we identified that using 1.0 equiv of 1, 1.0 equiv of

本卡为 NCBI PMC 开放获取 XML
的逐字摘录（标题、作者、表格行列、脚注、正文句均未改写），用于参照集候选与覆盖面登记；未逐条人工核验，标记
imported_unchecked，不写入判定记录、不参与 ChemCritic 引擎的可比性判定。对外引用一律回到 DOI 原文核对页码与表号。